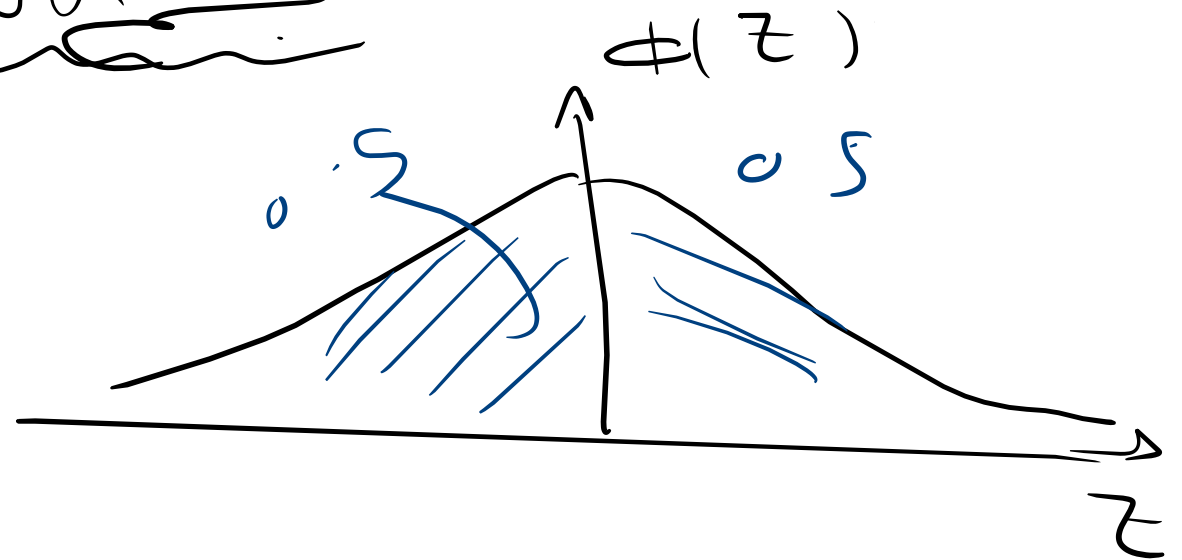


S.N.D.

*Handwritten scribbles*

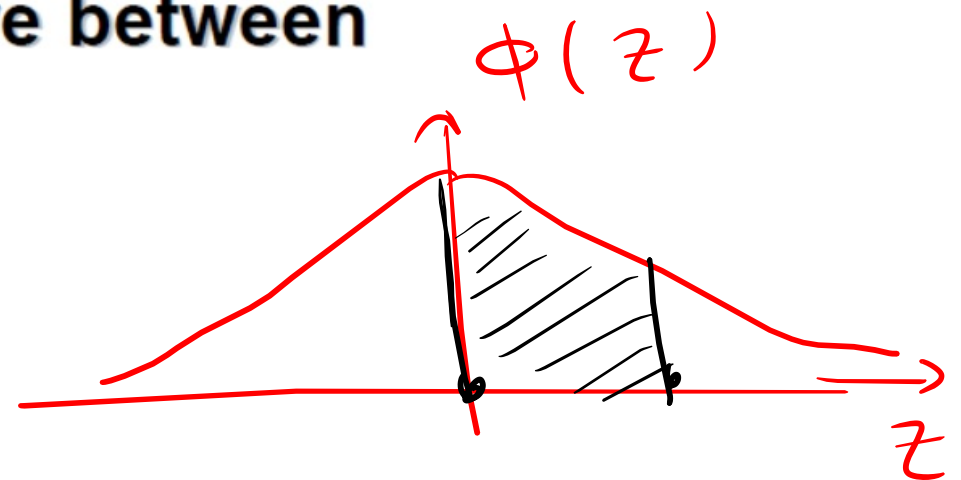


$$P(Z < 1.34) = 0.9099$$



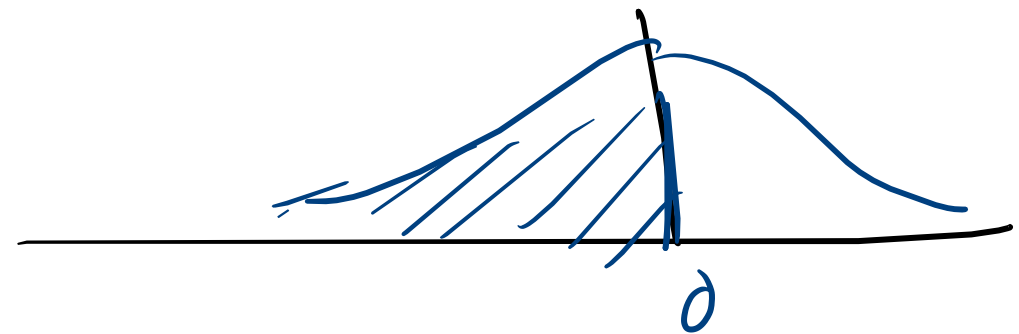
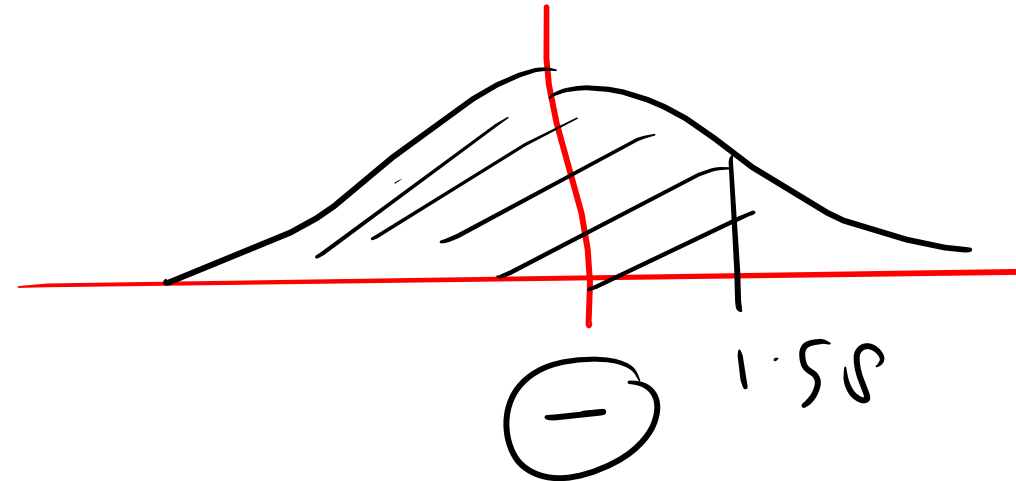
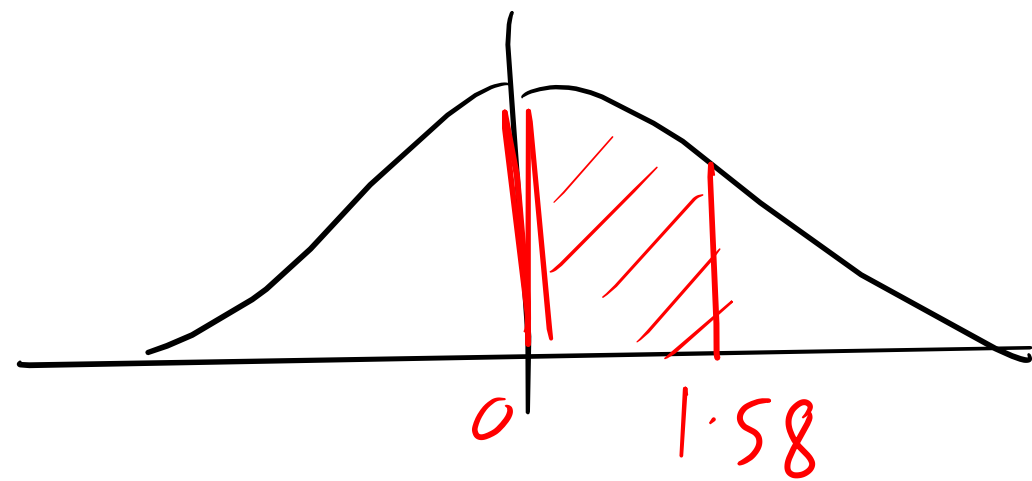
## ● ● ● | Example 1

- Find the area under the S.N.D. curve between  $z = 0$  and  $z = 1.58$ .

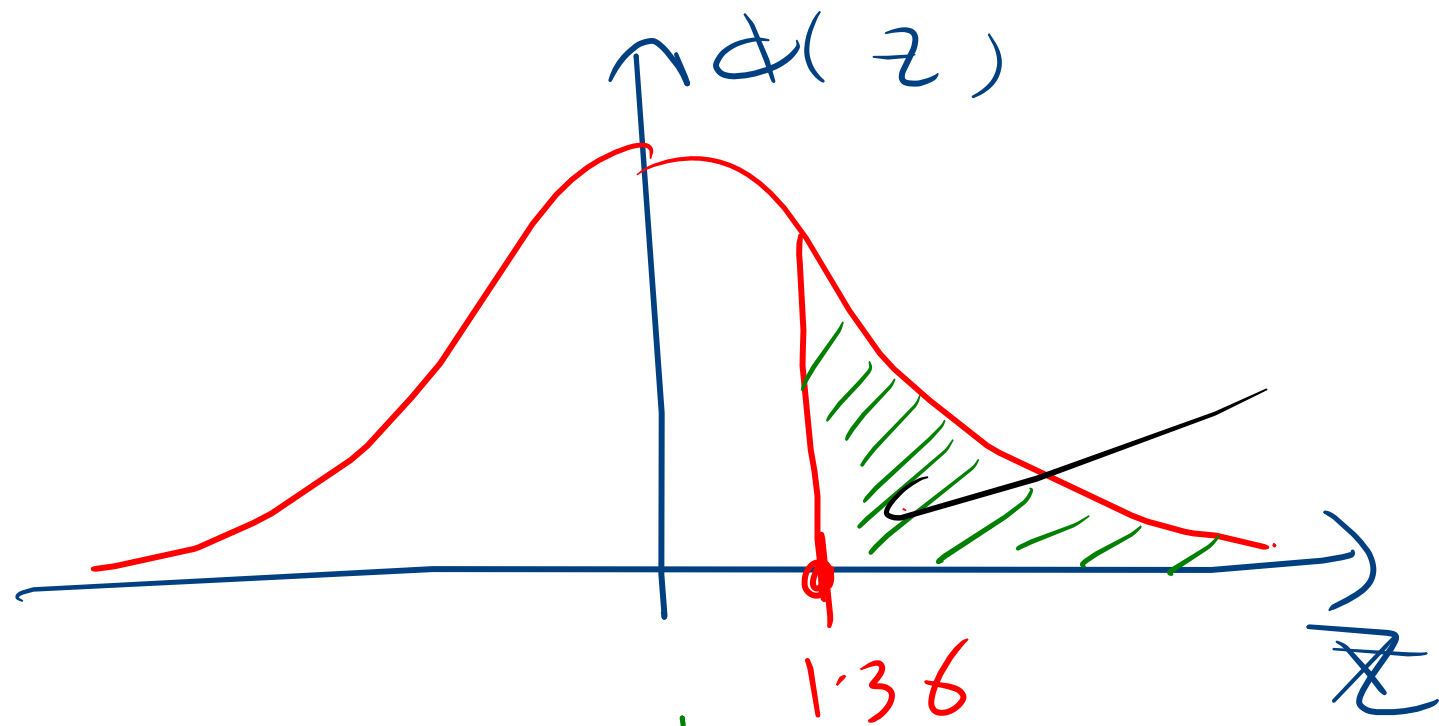


$$\begin{aligned}
 P(0 < Z < 1.58) \\
 &= \Phi(1.58) - \Phi(0) \\
 &= 0.9429 - 0.5
 \end{aligned}$$

✓



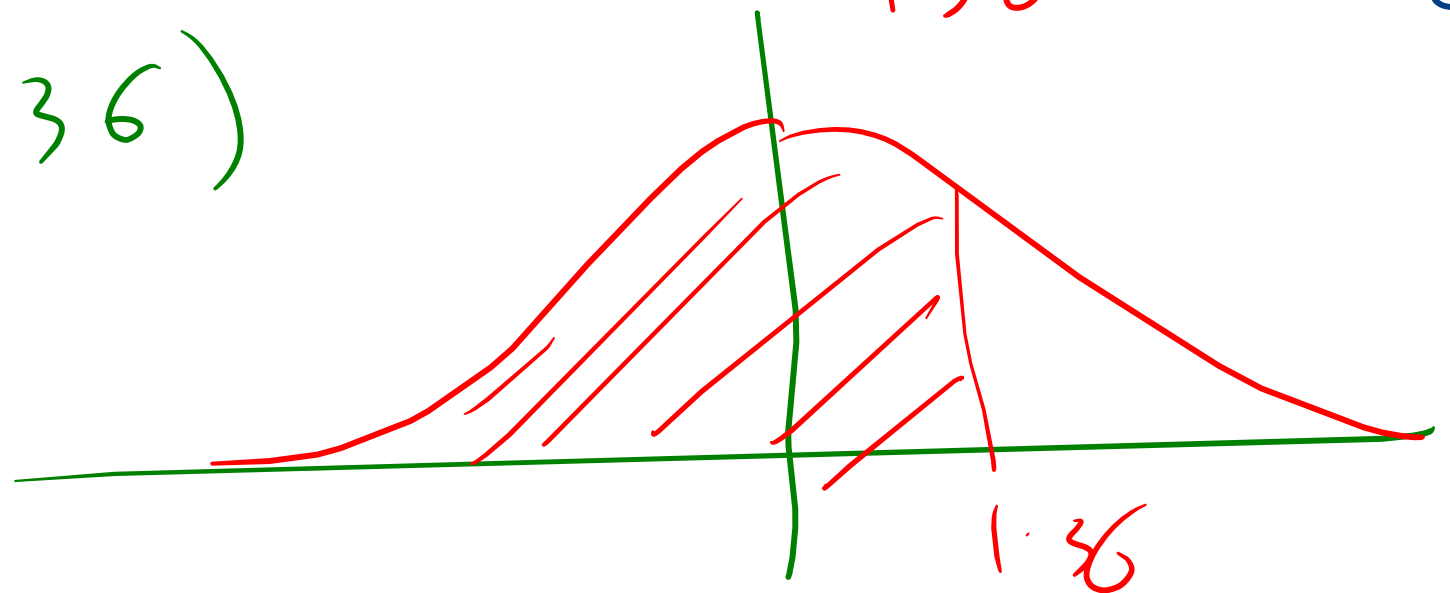
$$P(Z > 1.36)$$



$$= 1 - \Phi(1.36)$$

$$= 1 - 0.9131$$

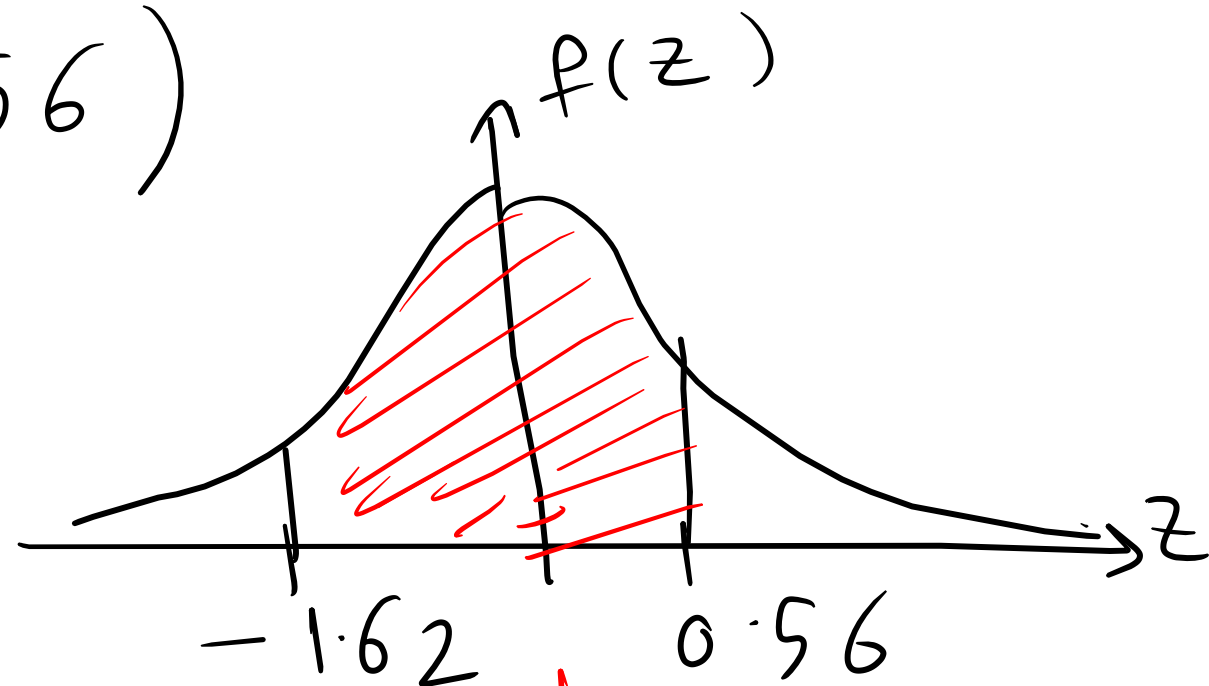
$$= 0.0869$$



$$P(-1.62 < Z < 0.56)$$

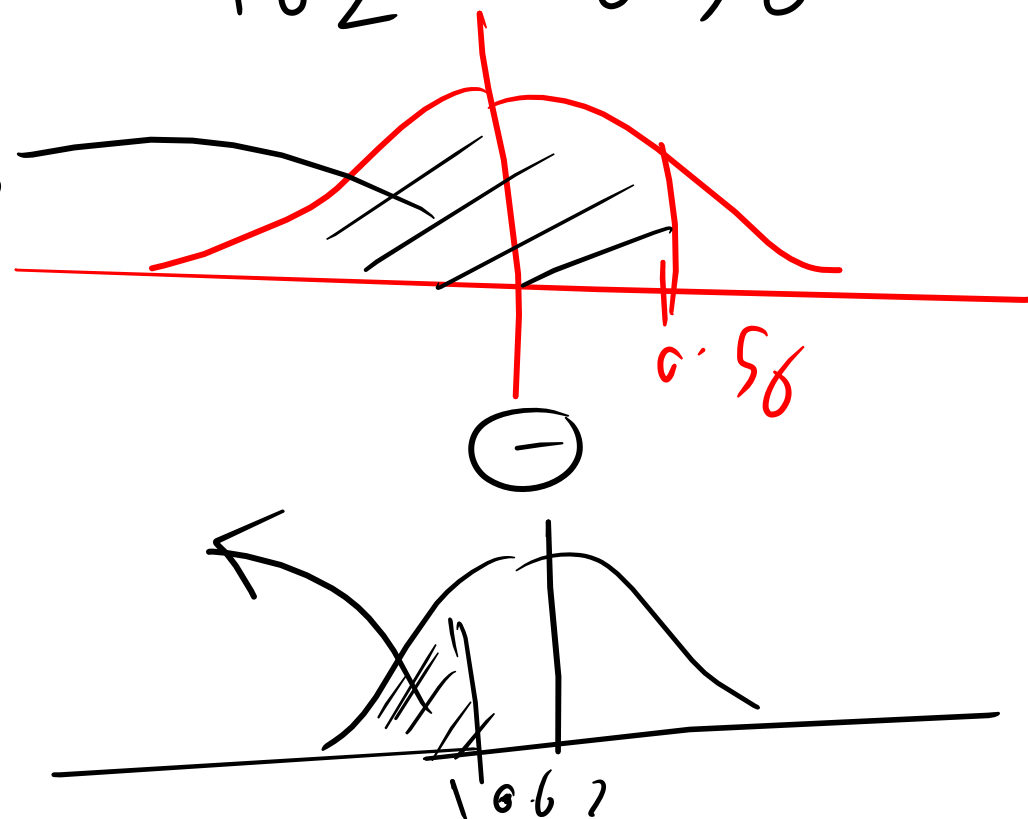
$$= \Phi(0.56) - \Phi(-1.62)$$

✗

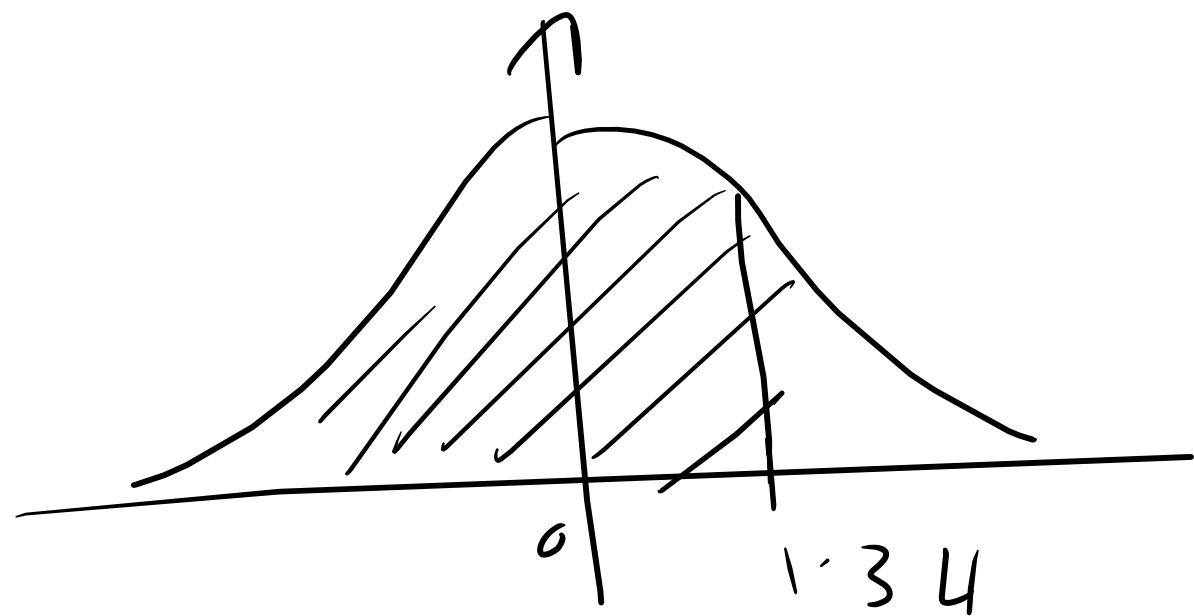


0.7183

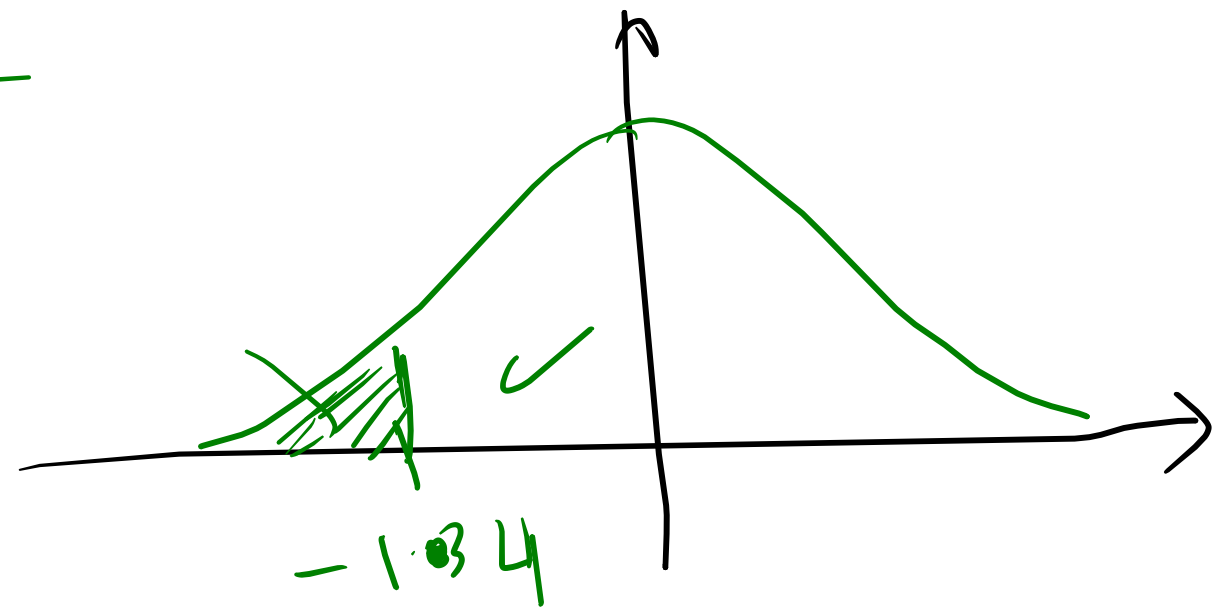
= 0.7123 -



$$\Phi(1.34) = ??$$



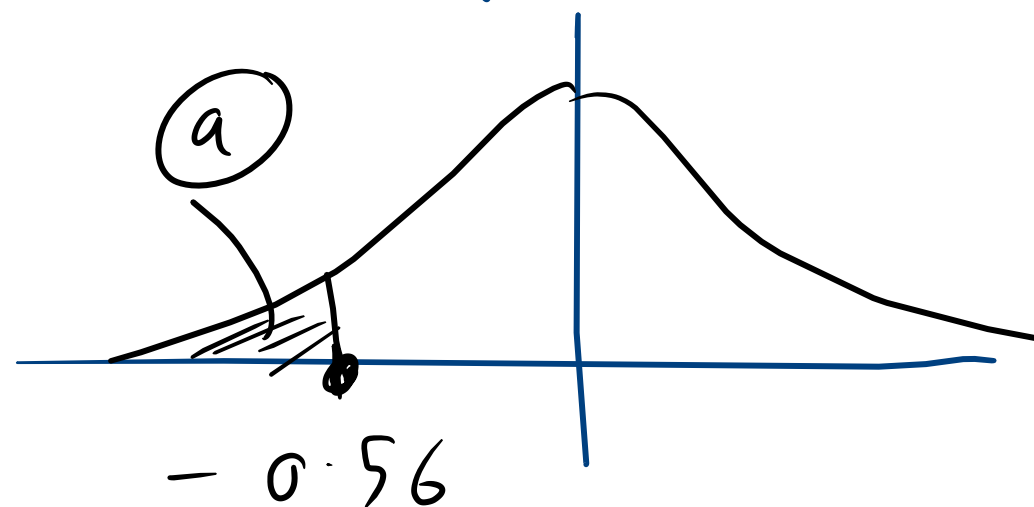
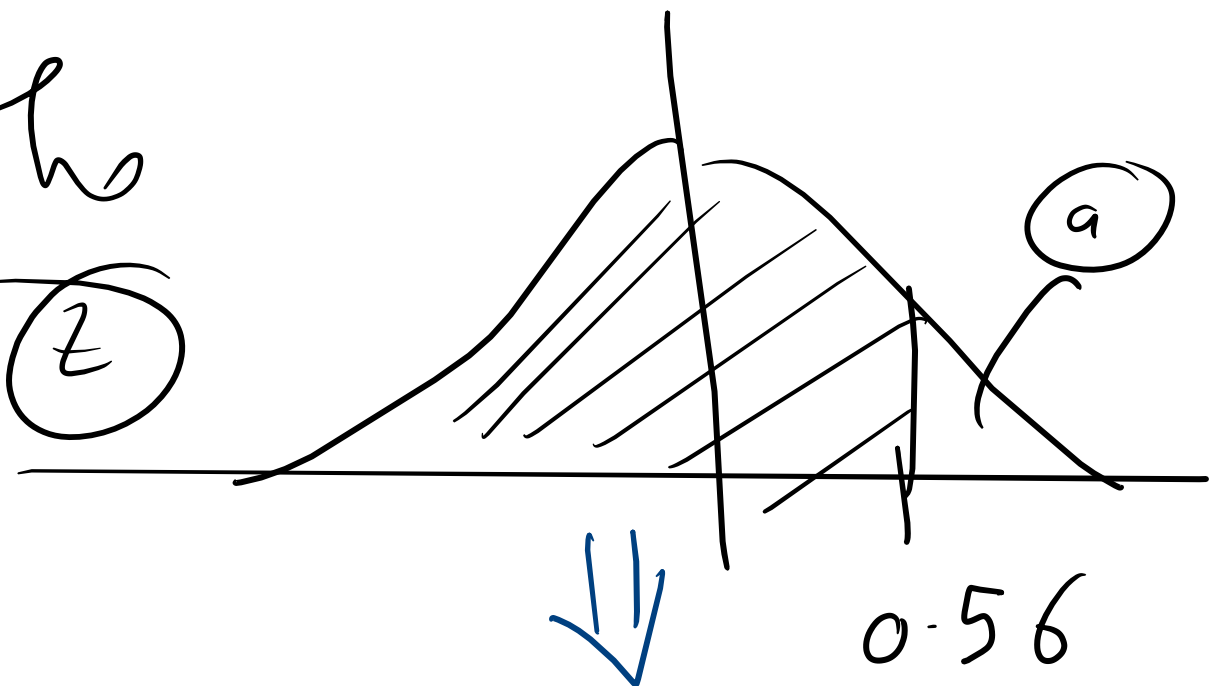
$$\Phi(-1.34) = \checkmark$$



$$= 1 - \Phi(-1.34)$$

I have only the  
Negative table of  $\Phi(z)$

$$\begin{aligned} &= 1 - \Phi(-0.56) \\ &= 1 - 0.2877 \\ &= 0.7123 \end{aligned}$$



$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left( \frac{x-\mu}{\sigma} \right)^2} dx$$

$$z = \frac{x-\mu}{\sigma} \Rightarrow \int_a^b \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left( \frac{x-\mu}{\sigma} \right)^2} dx$$

new limits for new Assumptions

$$\int_a^b \frac{1}{\sigma \sqrt{2\pi}} e^{-\left(\frac{x-\mu}{\sigma}\right)^2 \frac{1}{2}} dx$$

$$Z = \frac{x - \mu}{\sigma} \Rightarrow x = \sigma Z + \mu$$

at  $x = a$

$$Z = \frac{a - \mu}{\sigma}$$

at  $x = b$

$$Z = \frac{b - \mu}{\sigma}$$

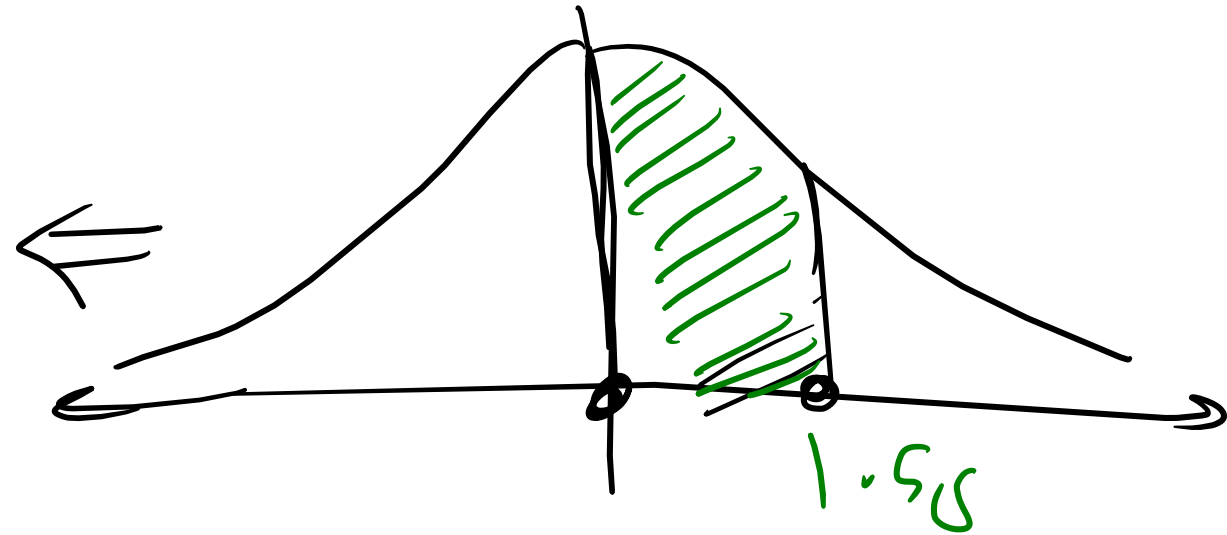
$$\int \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \quad \left| \begin{array}{l} x = \sigma z + \mu \\ dx = \sigma dz \\ dx = \sigma dz \end{array} \right.$$

$$= \frac{1}{\sigma \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz$$

- Find the area under the S.N.D. curve between  $z = 0$  and  $z = 1.58$ .



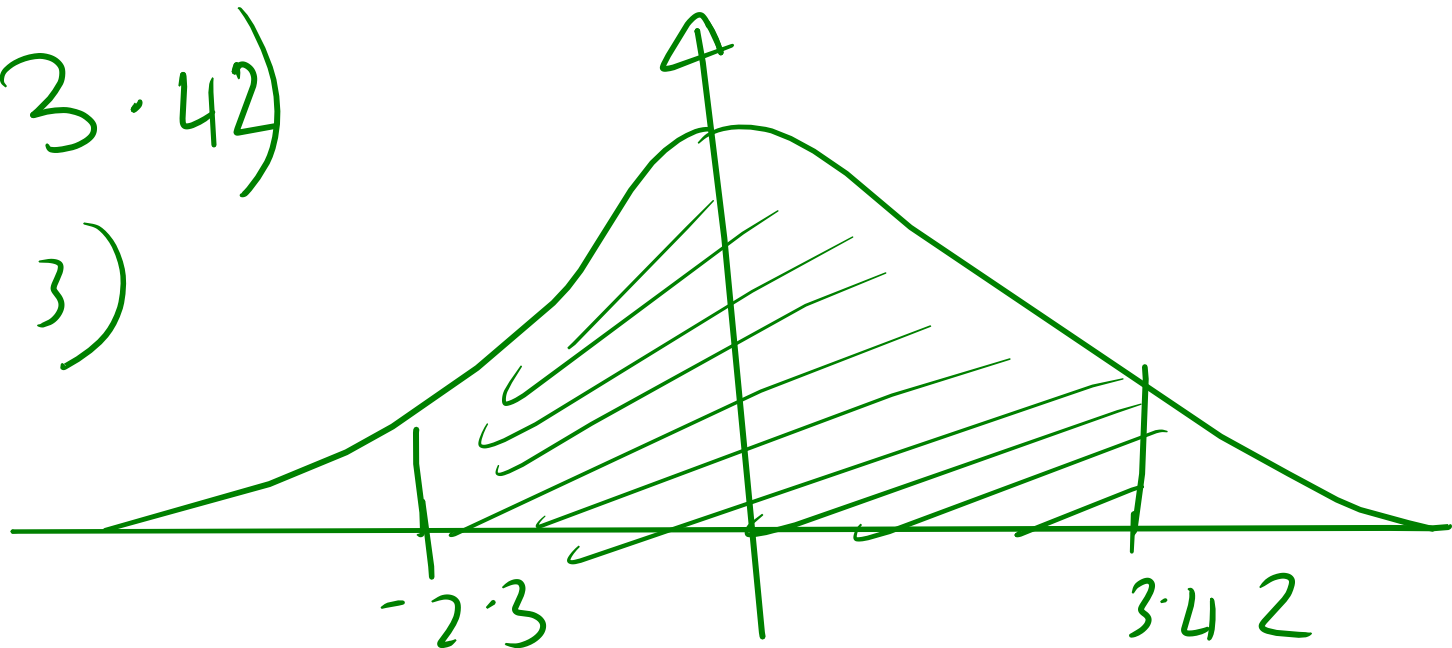
$$\begin{aligned} &= \Phi(1.58) - \Phi(0) \\ &= 0.9429 - 0.5 \\ &= 0.4429 \end{aligned}$$



$$P(-2.3 < Z < 3.42)$$

$$= \Phi(3.42) - \Phi(-2.3)$$

↓



$$= 0.9997 - 0.0107 =$$

$$= 0.9890$$

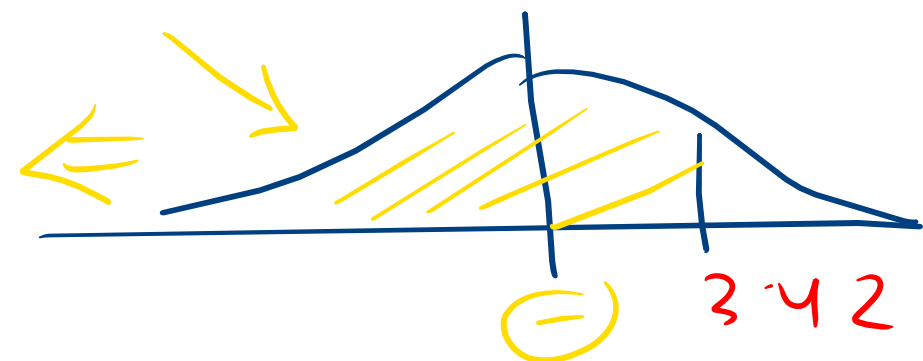
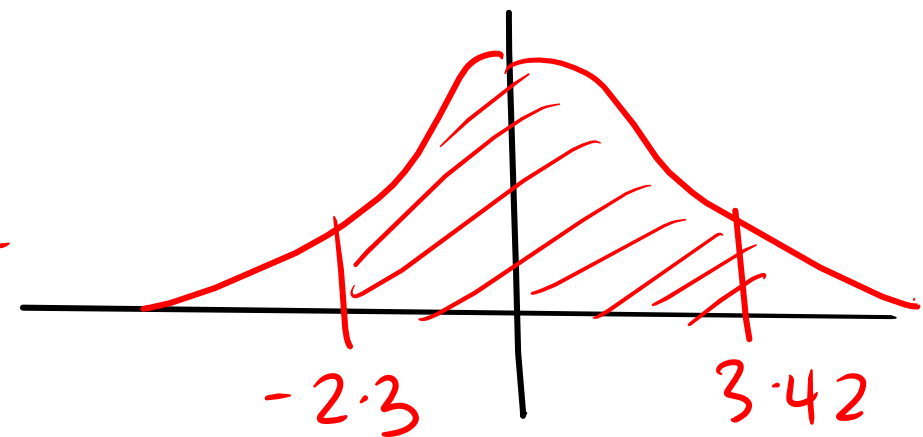
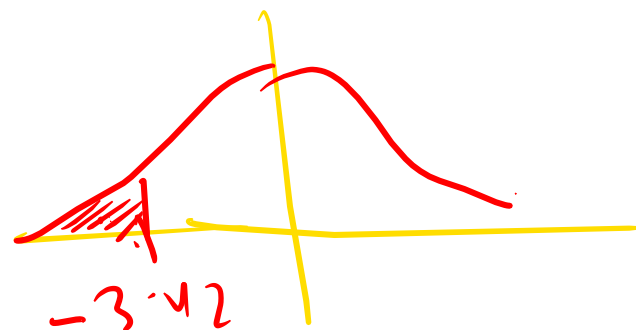
$P(-2.3 < Z < 3.42)$   
 using Z-negation table only

$$= \Phi(+3.42) - \Phi(-2.3)$$

$\downarrow$  from table  $\leftarrow$

$$(1 - \Phi(-3.42))$$

$$= 1 - \Phi(-3.42) - \Phi(-2.3)$$



$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$\mu = \checkmark$$

$$\sigma = \checkmark$$

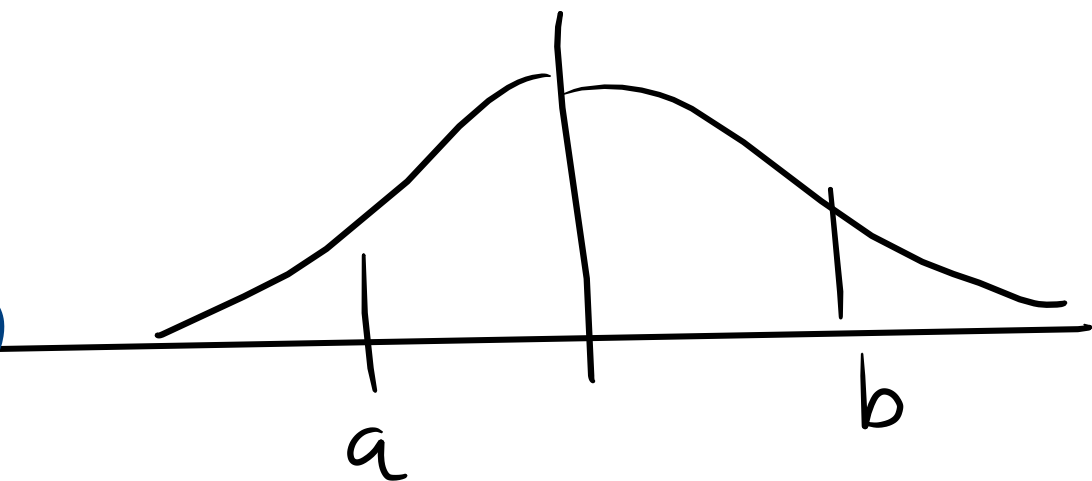
$$\mu = 0 \quad \sigma = 1 \Rightarrow$$

Standard Normal  
distribution

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2} dx$$

$$\int_a^b \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left( \frac{x - \mu}{\sigma} \right)^2} dx$$



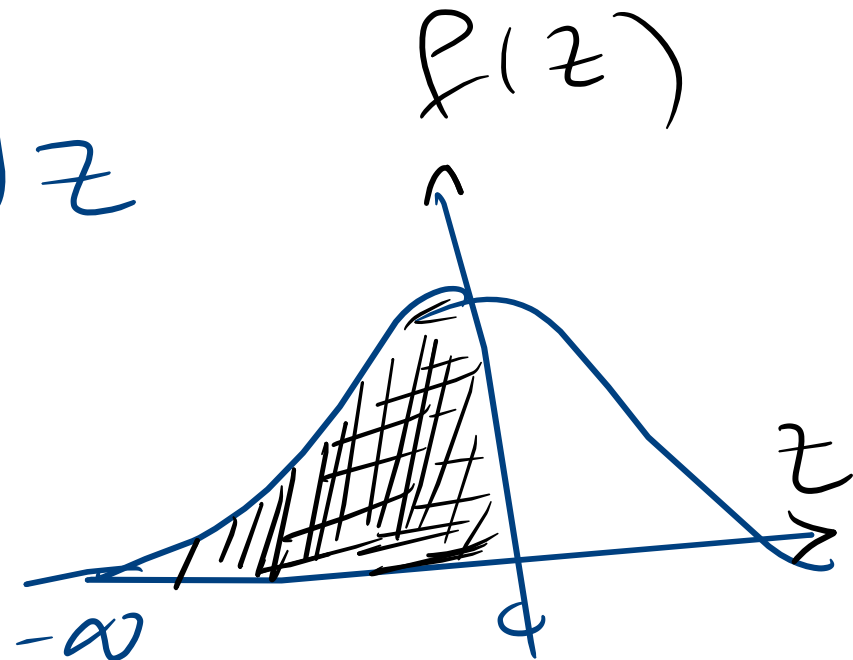
$$z = \frac{x - \mu}{\sigma} \Rightarrow \boxed{x = \sigma z + \mu}$$

$$dx = \sigma dz + 0$$

new limit

$$\int_{z_a}^{z_b} \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz$$

$$p(z) = \frac{1}{\sqrt{2\pi}} \int_{z_a}^{z_b} e^{-\frac{z^2}{2}} dz$$



$$p(z) = \int_{-\infty}^0 \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz =$$

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt$$

$$\Gamma(n) = \int_0^{\infty} e^{-t} t^{n-1} dt$$

$$\Gamma(3) = \int_0^{\infty} e^{-t} t^2 dt$$

$$I = \int_{-\infty}^0 e^{-\frac{z^2}{2}} dz \rightarrow \text{?????}$$

let  $\frac{z^2}{2} = t \Rightarrow z^2 = 2t \Rightarrow z = \sqrt{2t}$   
 $dz = \frac{2}{2\sqrt{2t}} dt$

$$dz = \frac{1}{\sqrt{2t}} dt$$

$$\boxed{dz = \frac{1}{\sqrt{2t}} dt}$$

$$-\int_{\infty}^0 \square dz$$

$$z^2 = 2t \Rightarrow t = \frac{z^2}{2} \Rightarrow \begin{matrix} dz \\ \therefore \end{matrix} \begin{matrix} z = -\infty \\ \boxed{t = \infty} \end{matrix}$$

$$\text{at } z = 0 \Rightarrow t = \frac{0}{2} = 0$$

$$\int_{\infty}^0 e^{-t} \frac{1}{\sqrt{2t}} dt = \frac{1}{\sqrt{2}} \int_{\infty}^0 t^{-1/2} e^{-t} dt$$

$$= \frac{1}{\sqrt{2}} \oint_C e^{-t} t^{-1/2} dt$$

$$n-1 = -\frac{1}{2}$$

$$\Rightarrow \boxed{n = \frac{1}{2}}$$

$$= \frac{1}{\sqrt{2}} \Gamma\left(\frac{1}{2}\right) = \frac{\sqrt{\pi}}{\sqrt{2}}$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\int_{-\infty}^0 f(x) dx = - \int_0^{\infty} f(x) dx$$

**4-39.** Use Appendix Table II to determine the following probabilities for the standard normal random variable  $Z$ :

- (a)  $P(Z < 1.32)$
- (b)  $P(Z < 3.0)$
- (c)  $P(Z > 1.45)$
- (d)  $P(Z > -2.15)$
- (e)  $P(-2.34 < Z < 1.76)$

a)  $P(Z < 1.32) = 0.90658$

b)  $P(Z < 3.0) = 0.99865$

c)  $P(Z > 1.45) = 1 - 0.92647 = 0.07353$

d)  $P(Z > -2.15) = P(Z < 2.15) = 0.98422$

e)  $P(-2.34 < Z < 1.76) = P(Z < 1.76) - P(Z > 2.34) = 0.95116$

**4-47.** The compressive strength of samples of cement can be modeled by a normal distribution with a mean of 6000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter.

- (a) What is the probability that a sample's strength is less than 6250 Kg/cm<sup>2</sup>?
- (b) What is the probability that a sample's strength is between 5800 and 5900 Kg/cm<sup>2</sup>?
- (c) What strength is exceeded by 95% of the samples?

$$\begin{aligned} \text{a) } P(X < 6250) &= P\left(Z < \frac{6250 - 6000}{100}\right) \\ &= P(Z < 2.5) \\ &= 0.99379 \end{aligned}$$

$$\begin{aligned} \text{b) } P(5800 < X < 5900) &= P\left(\frac{5800 - 6000}{100} < Z < \frac{5900 - 6000}{100}\right) \\ &= P(-2 < Z < -1) \\ &= P(Z < -1) - P(Z < -2) \\ &= 0.13591 \end{aligned}$$

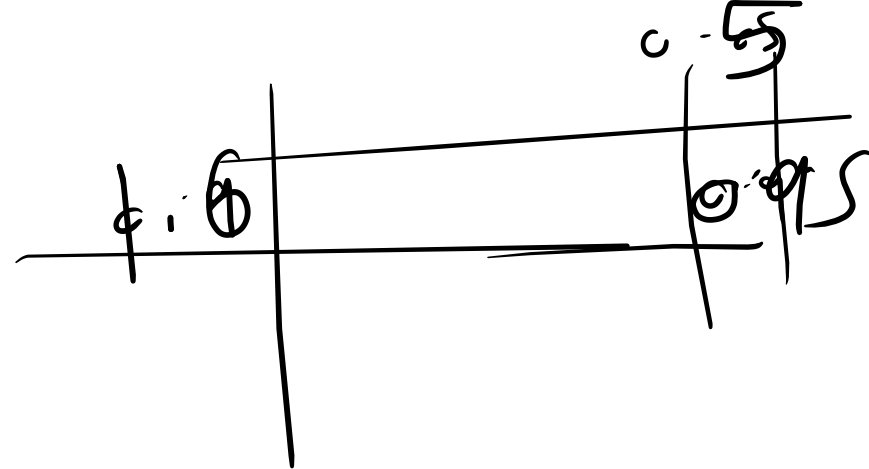
$$\text{c) } P(X > x) = P\left(Z > \frac{x - 6000}{100}\right) = 0.95.$$

$$\text{Therefore, } \frac{x - 6000}{100} = -1.65 \text{ and } x = 5835.$$

$$P(Z \leq \frac{X - 6000}{1000}) = 0.95$$

from table

1.65



$$\frac{X - 6000}{1000} = -1.65$$

$$\therefore X = 4350$$

